BIOSTAT 707 Project Proposal #4 Challenge 2012

Group Members:

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Repository URL: https://github.com/jailegins/BIOSTAT707-Group-Project.git

Data Source: The PhysioNet/CinC Challenge 2012

Roles:

- **Tanaiya Sumter – Data Preparation:** Organize and clean the data, summarize the different ICU groups, and help prepare the data for analysis.
- **Tatyana Bowers – Model Development:** Help build and test the models to see how well they predict mortality when applied to a different ICU type.
- **Jai Legins – Model Comparison:** Compare how well the models perform and create tables or graphs to clearly present the results.

All group members will help make analysis decisions, interpret results, write the final report, and prepare the presentation.

This project aims to explore whether a model that utilizes data from a subset of intensive care unit types (i.e coronary care, cardiac surgery recovery, medical, surgical) will continue to perform well on the ICU type that is not used to create the model for predicting mortality of ICU patients. Specifically, we want to know if we create a model that only includes data from 3 of the 4 ICU types, will it perform well if tested on the 4th unused ICU type.

Approach:

We will create 4 models, and each model will be trained using data from 3 ICU types only. We will then test on the held-out ICU type and evaluate model performance. For example, Model 1 will be trained on coronary care, cardiac surgery recovery, and medical. Model 1 will then test on data introduced from surgical ICU. We will determine if the models

performance differs by comparing each held-out model to the other 3 models and a full model as a reference. Discrimination and calibration will be reported separately, and we will also try a simple recalibration (refitting the intercept and slope) on part of the held-out type and show how much it recovers.

Hypotheses:

We hypothesize that a model that trains using 3 ICU types only and testing on the held-out type will perform poorly in predicting mortality in ICU patients. Specifically, the 4 held-out models will not perform as well on predicting mortality as a model that includes all 4 will. We also hypothesize that a model that included coronary care or cardiac surgery while the other is held-out will perform better than a model that has both coronary care and cardiac surgery in the training data.